UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Engineering, Built Environment and IT
Department of Computer Science

COS791

Image Analysis and Understanding

Multilevel Thresholding

Due: 08 October 2026

Instructions

1. This is a group assignment.
2. Plagiarism is not allowed.
3. The use of libraries is permitted. You may use any programming language of your choice.
4. The assignment consists of 1 task.
5. The assignment carries a total of 50 marks.

Summary

Assignment Overview: Multilevel image thresholding is a core technique in digital image processing used to segment an image into multiple distinct regions based on pixel intensity distributions. However, as the number of threshold values K increases, finding the optimal set of thresholds using exhaustive search becomes computationally intractable due to exponential time complexity. Deep learning requires thousands of manually annotated, pixel-level ground truth masks, which are extremely expensive and time-consuming to produce (especially in medical imaging with expert radiologist annotations). Multilevel thresholding is completely unsupervised—it requires zero training data, zero manual labels, and zero offline training time.

In this assignment, you will implement, analyse, and empirically evaluate Differential Evolution (DE) and its variants to solve the multilevel image thresholding optimisation problem using Otsu's, Kapur's Entropy and Tsallis as objective functions.

Your experiments must evaluate performance across two distinct datasets:

1. **Phase 1 (Proof-of-Concept / Standard Benchmarks):** A dataset of 10 standard (BDS500) test images to validate your algorithmic implementation and establish initial baseline convergence behaviours.
2. **Phase 2 (Domain Application / Medical Imaging):** A dataset of 15 CHAOS MRI scan images to assess algorithm performance, robustness, and stability on real-world medical data.

1 Algorithmic Framework & Problem Formulation

Let an image I of size $M \times N$ have L gray levels $\{0, 1, \dots, L-1\}$. The normalised histogram probability distribution is given by $p_i = h(i)/(M \times N)$, where $h(i)$ is the frequency of intensity level i.

Given K threshold values $\mathbf{t} = [t_1, t_2, \dots, t_K]$ such that $0 < t_1 < t_2 < \cdots < t_K < L-1$, the image is partitioned into $K+1$ intensity classes $C_0, C_1, \dots, C_K$.

1.1 Threshold Levels to Evaluate

You must evaluate all algorithms across six distinct threshold levels:

$$K \in \{3, 5, 7, 9, 11, 12\} \tag{1}$$

1.2 Objective Functions (Optimisation Targets)

You will use each of the following three objective functions:
Consult literature for implementation details.

1. Otsu's Variance (f_{Otsu})
2. Kapur's Information Entropy (f_{Kapur})
3. Tsallis Non-Extensive Entropy (f_{Tsallis})

1.3 Optimization Algorithms to Implement

You must implement and compare the following **six DE variants**:
Consult literature for implementation details.

1. **Standard DE (DE/rand/1/bin):** Baseline classical Differential Evolution.
2. **JADE:** Adaptive DE with optional external archive and p-best mutation strategy (`DE/current-to-pbest/1`).
3. **SHADE:** Success-History based Adaptive DE using historical memory for F and CR parameter control.
4. **L-SHADE:** SHADE integrated with Linear Population Size Reduction (LPSR) over the runtime budget.
5. **Late Acceptance DE** - adopted from Late Acceptance Hill Climbing.

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2 Performance Evaluation Metrics

2.1 Image Reconstruction Metrics

1. **Peak Signal-to-Noise Ratio (PSNR):**
2. **Structural Similarity Index (SSIM):**
3. **Feature Uniformity Metric (U):**

3 Experimental Protocols & Parameters

- **Number of Independent Runs:** Perform **30 independent runs** per algorithm for every combination of (Image, Objective Function, K).
- **Stopping Criteria:** Set maximum Function Evaluations (FEs) to be equal.
- **Statistical Significance:** Conduct **Wilcoxon signed-rank tests** ($\alpha = 0.05$) or **Friedman rank tests** to establish statistically significant superiority among the algorithms.

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4 Submission & Deliverables

Your final submission must include:

1. **Source Code Repository:** Self-contained, clean, and documented code. A master script must reproduce all experimental results and save output tables/plots.
2. **Technical Report (PDF):** Structured in Expert Systems With Application (ESWA) journal :
   - Methodology & Algorithmic parameter settings.
   - Tabulated Mean & Standard Deviation values for PSNR, SSIM, and Uniformity across $K \in \{3, 5, 7, 9, 11, 12\}$.
   - Convergence curves comparing DE, JADE, SHADE, L-SHADE and LADE.
   - Visual segmentation outputs displaying original vs. segmented images.
   - Critical analysis discussing execution time, algorithm scalability, and the impact of non-extensive entropy (q) vs. Otsu/Kapur.

Table 1: Comparison of PSNR ($\mu \pm \sigma$) for CHAOS MRI Slices at $K = 12$ over 30 runs.

Algorithm	**Otsu**	**Kapur**	**Tsallis** ($q = 0.8$)
Standard DE	28.45 ± 1.23	27.12 ± 1.85	26.90 ± 1.95
JADE	31.10 ± 0.42	30.05 ± 0.61	29.80 ± 0.74
SHADE	32.85 ± 0.15	31.90 ± 0.22	31.50 ± 0.28
L-SHADE	$\mathbf{33.40 \pm 0.04}$	$\mathbf{32.65 \pm 0.08}$	$\mathbf{32.10 \pm 0.11}$
CoDE	30.50 ± 0.88	29.40 ± 1.10	28.95 ± 1.20
SaDE	31.95 ± 0.31	30.80 ± 0.45	30.25 ± 0.50

5 Mark Allocation (Total: 50 Marks)

Section	**Description**	**Marks**
1. Implementation & Framework	Correct implementation of the 5 DE algorithms (DE, JADE, SHADE, L-SHADE, LADE) and the 3 objective functions (Otsu, Kapur, Tsallis).	**10**
2. Experiment 1	Execution, tabulation, and analysis of reconstruction metrics (PSNR, SSIM, Uniformity) across all K levels on BSD500.	**10**
3. Experiment 2	Domain-specific application, GT mask matching procedure, evaluation of Jaccard and Dice metrics on CHAOS MRI.	**15**
4. Statistical & Scalability Analysis	Rigorous statistical testing (Wilcoxon/Friedman), convergence rate plots, computational time vs. K scalability analysis.	**5**
5. Report & Code Quality	Professional presentation, adherence to ESWA format, reproducible code structure, clear code documentation.	**10**
Reporting template https://www.overleaf.com/latex/templates/eswa-journal-article-template/xryvqrgpxdvx **TOTAL**		**50**